

ASSESSMENT TOOL 1

Concept Mapping

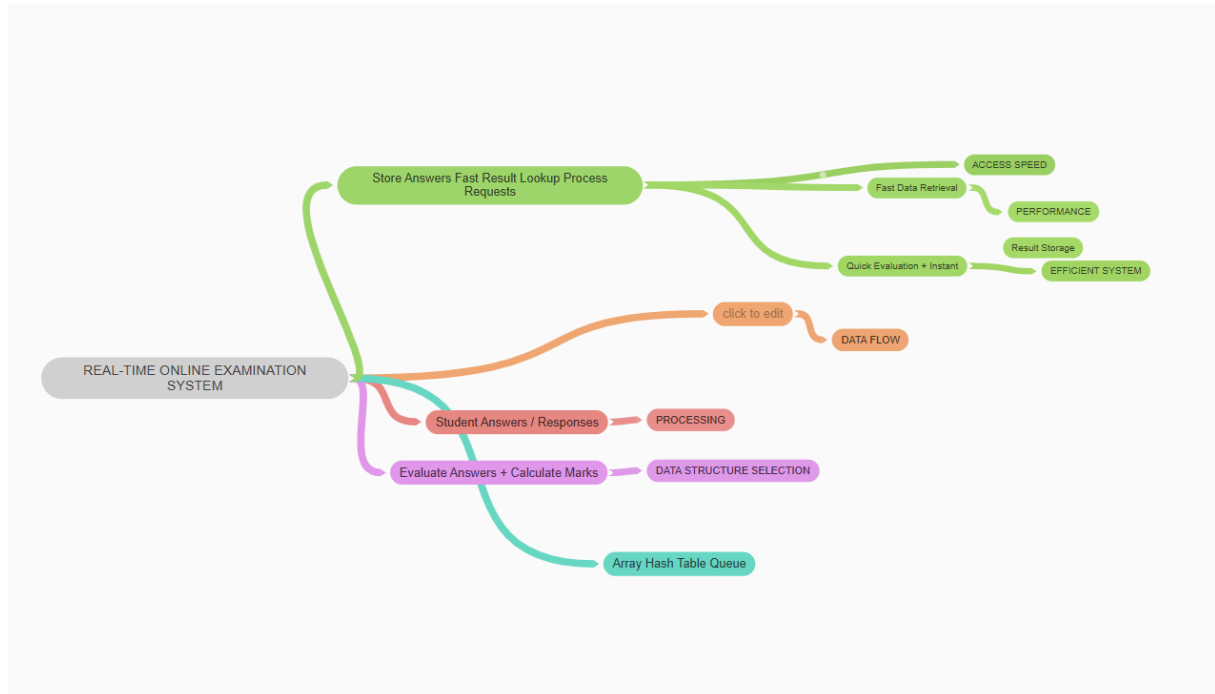
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Course Code: CSA0302

Course Name: Data Structures

Q32. A real-time online examination system processes student responses, evaluates answers, and stores results instantly. Construct a concept map linking Data Flow → Processing → Data Structure Selection → Access Speed → Performance. Analyze how the flow of data influences the choice of data structures and justify the most efficient design.



Q33. A search engine must return results quickly while handling massive datasets. Construct a concept map linking Speed → Memory Usage → Time Complexity → Space Complexity → Trade-Off → Optimization. Analyze the trade-offs and justify how balance is achieved.

